Project Specification and Plans

CITS5206 INFORMATION TECHNOLOGY CAPSTONE PROJECT

GROUP 11

DELIVERABLE 1

CITS5206 Group 11

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1. Problem statement

1.1 Project Background

GJP Engineering proposed the **AssetGuard AI** project to support asset managers and consulting engineers in processing structural load requests. In practical scenarios, maintenance staff or contractors may need to place equipment or machinery on existing structures such as bridges, platforms, or building areas. In these cases, decision-makers need to determine whether the proposed load complies with the structural limits specified in engineering drawings and design criteria. According to the client's kickoff materials, the system is intended to assist users in assessing these types of load requests.

At present, this process mainly relies on manually reviewing and interpreting engineering drawings. During the meeting, the client noted that users may sometimes find it difficult to identify the latest version of a drawing and may not always be able to accurately interpret the key information it contains. This increases the time required to handle requests and also raises the risk of incorrect judgement, especially for users without an engineering background. The project therefore aims to provide a clearer system workflow to support load input, structural limit comparison, and result evaluation.

1.2 Current Workflow

At present, structural load requests are primarily assessed through manual review of engineering drawings. When maintenance staff or contractors propose placing equipment on existing structures (such as bridges, platforms, or buildings), responsible personnel must first locate the relevant drawings, verify their validity, and then identify the applicable design parameters and structural load limits before making a decision.

This process relies heavily on engineering expertise and individual interpretation. Users are required to determine whether the drawings are up to date and accurately extract key structural information, which can be challenging in practice. As noted by the Client, it is not always straightforward to identify the latest version of a drawing or to correctly interpret the critical data it contains.

As a result, the current workflow can be time-consuming, difficult for users without an engineering background, and prone to inconsistency or error in decision-making.

1.3 Problems in the Current Process

Although the current workflow allows staff to assess structural load requests by referring to engineering drawings, it still has several limitations in practice. Based on the Client's meeting discussion and the current project context, the main problems can be summarized as follows:

- **The assessment process relies too heavily on manual drawing review**
  At present, when handling structural load requests, staff need to locate the relevant engineering drawings and make decisions based on the design parameters and load limits shown in those drawings. This process is completed manually, involves multiple steps, and is relatively inefficient.
- **Drawing versions and key information are not easy to confirm**
  During the meeting, the client mentioned that users may sometimes find it difficult to confirm whether they are using the latest version of a drawing, and they may not always be able to quickly identify the key information related to structural limits.
- **The process depends strongly on engineering knowledge**
  The current workflow requires users to understand structural design information and load parameters shown in engineering drawings. As a result, it is not easy to use for people without an engineering background.
- **Assessment results may become inconsistent**
  Since drawing interpretation and load assessment mainly depend on human understanding, different staff members may interpret the same request differently, which can affect the consistency of the final result.
- **The overall process is relatively slow**
  From locating the drawings and confirming the relevant information to making the final assessment, the whole process can take time and may not support timely responses to structural load requests in practice.

1.4 Rationale for the Proposed System

Based on the limitations of the current workflow, there is a clear need for a more structured and consistent system. At present, structural load requests are mainly assessed through manual review and interpretation of engineering drawings. This not only makes the process slower but also places a high demand on users' technical knowledge and personal judgement. During the meeting, the client noted that users may sometimes find it difficult to identify the latest version of a drawing and may not always be able to accurately interpret the key information it contains. As a result, the current process is difficult to carry out efficiently and consistently in practice.

The proposed system is intended to make the handling of load requests clearer and easier to manage. Users would be able to select the relevant structure, enter the proposed equipment load, and compare it against the available structural limit information within the system. For the client, such a system would reduce the burden of manually checking drawings, extracting information, and making repeated judgements. It would also make the process easier to use for people without an engineering background. Based on the kickoff materials and meeting discussion, the purpose of the system is not to replace engineering judgement itself, but to provide a more standardized and accessible way to handle load requests.

In addition, the current manual process has limited traceability. Manual assessments often do not provide a complete record of what information was used, how the decision was made, or who made it. From a compliance and audit perspective, this makes it difficult to provide clear records. In particular, if a safety incident occurs, the assessment process may be difficult to use as complete compliance evidence. By comparison, a digital system can improve traceability by providing a more consistent and recordable workflow.

In practical asset management work, these requests often need to be handled in a timely manner. If the process continues to rely entirely on manually checking drawings, confirming parameters, and making decisions, it is likely to remain slow and more vulnerable to inconsistent outcomes. By introducing a dedicated system, the client aims to make this process more direct and easier to use, while also improving efficiency, consistency, and traceability in day-to-day work.

2. Client communication and MVP agreement

2.1 Client Communication Methodology

To ensure effective collaboration with the client and maintain alignment throughout the project lifecycle, our team established structured communication channels, clearly defined stakeholder roles, and a consistent feedback process based on ongoing client meetings and iterative updates.

Stakeholders

- Client Representative: GJP Engineering (Structural Engineering Team)
- Key Technical Contacts:
    - Ryan Morrison (UI / Web Expertise)
    - Tahmer Hijjawi (Senior Software Engineer)
    - Kelvin Hands (Systems / Autonomous Engineering)
- Development Team: UWA Master of IT Students (Group 11 Tean 3a)

Communication Channels

- **Microsoft Teams:**
  Used for weekly meetings, discussions, and coordination.
- **Email (Outlook):**
  Used for formal communication and requirement confirmation (client contact: glen@gjpengineering.com.au).
- **GitHub:**
  Used for accessing the existing AI system, tracking development progress, and managing tasks.

- **OneDrive / SharePoint:**
  Used for sharing project data, documentation, and maintaining communication records.

Communication Approach

Our communication approach follows an iterative and client-centered workflow:

- Weekly in-person meetings are conducted to provide updates and receive feedback
- Key decisions and confirmations are documented via email
- Teams is used for quick discussions and coordination
- GitHub ensures transparency in development progress

Additionally, the team maintains a Request for Information (RFI) register to track unclear requirements and questions raised during meetings.

Weekly In-person Meeting Plan

WEEK	DATE	ACTIVITY	METHOD	PURPOSE
Week 3	11 Mar	Initial Client Meeting	In-person	Understand project background and requirements
Week 4	18 Mar	Requirement Clarification	In-person	Confirm system scope and MVP direction
Week 5	25 Mar	System Design Discussion	In-person	Present system workflow
Week 6	1 Apr	Development	In-person	Finalize MVP features
Week 7-12	Ongoing	In progress	In-person	Iterative feedback and improvement

2.2 Client Requirement Understanding

Our understanding of the client requirements was developed through a structured communication process consisting of three key stages: initial requirement gathering, proposal-based clarification, and final MVP confirmation.

Initial Client Requirements (Kickoff Meeting)

At the kickoff meeting, the client introduced the project background, business context, and expected system direction. The core requirement identified was to support asset managers and engineers in processing structural load requests more efficiently.

From the client’s description and case scenarios, the system is expected to:

- Provide a user-accessible platform for handling load requests
- Allow users to select assets and input proposed loads
- Compare the input load against structural limits extracted from engineering drawings
- Return a clear compliance result (pass/fail)

- Improve efficiency, reduce manual interpretation, and support better decision-making

In addition, the client introduced an existing AI-based drawing extraction tool and indicated that the system should integrate with this data source.

Proposal and Requirement Clarification

Based on the initial requirements, the team developed an MVP proposal and system workflow design. During this stage, several key uncertainties and assumptions were identified and raised to the client for clarification.

The main clarification points included:

- Definition of roles (e.g., Admin, Asset Manager, Contractor)
- Scope of the evaluation logic (simple comparison vs. advanced engineering rules)
- Whether email notifications should be automatic or user-triggered
- The level of complexity required for asset data and engineering constraints
- Input structure (equipment type, load values, and unit handling)

These questions were documented and discussed with the client through meetings and follow-up communication. This process ensured that assumptions made in the proposal were validated before implementation.

Final Agreed Understanding

After iterative discussions and clarification, the team and client reached a shared understanding of the system scope for the MVP stage.

The agreed system is a **decision-support tool** that:

- Allows users to log in and access the system
- Enables selection of assets and input of load information
- Performs a rule-based comparison between input load and structural limits
- Returns a clear compliance result (Compliant / Non-Compliant)
- Records evaluation results for traceability
- Supports optional (user-triggered) email reporting

Importantly, the system is not intended to replace engineering judgement, but to provide a structured, consistent, and user-friendly workflow to assist decision-making.

This staged communication process ensures that the current requirement understanding is directly grounded in client input, validated assumptions, and mutually agreed system scope.

2.3 Client Alignment on the MVP Scope (MVP Agreement)

The MVP scope was established through a progressive alignment process between the client's initial vision and the team's proposed solution.

From Vision to MVP Definition

During the kickoff meeting, the client outlined a high-level MVP vision, which included a relatively complete workflow with features such as user authentication, asset selection, load input, AI-assisted compliance checking, and automated notifications.

However, this vision represented a broader product direction rather than a strictly scoped MVP suitable for delivery within the project timeframe.

Refinement Through Proposal and Discussion

Based on this vision, the team proposed a simplified and achievable MVP focused on the core workflow. During this process, several design decisions were refined through client discussion:

- The compliance check will use basic rule-based comparison, rather than complex engineering modelling
- Email notifications will be user-triggered, not automatic
- The system will prioritise core workflow completeness over feature richness
- Advanced features (e.g., complex constraints, automation, full audit systems) will be deferred

This refinement ensured that the MVP remains technically feasible while still delivering meaningful value.

Final MVP Agreement

Following these discussions, the client and team agreed that the MVP will focus on the core load assessment workflow, including:

- **User Login and Basic Access**
  The system should provide a basic login function so that relevant users can access the platform and use the load request handling workflow. The main purpose of this feature is to support basic access control and ensure that the system operates with clearly identified users.
- **Structure Selection and Load Input**
  Users should be able to select the structure or asset to be assessed, such as a bridge, platform, or building area, and enter the proposed equipment type, planned load, and units. This function forms the core input for the assessment workflow and serves as the basis for subsequent structural limit comparison and result evaluation. By

integrating structure selection and load input into one workflow, the system can support a more direct submission process.

- **Structural Limit Comparison and Basic Assessment**
  The system should be able to compare the user-entered load against the available structural limit information and provide a clear assessment result. According to the meeting discussion, the focus at this stage is not to develop a complex engineering calculation model, but to implement a clear and executable comparison process that helps users determine whether the request satisfies the relevant limits. For this reason, this function represents the core of the MVP.
- **Result Display**
  After the comparison is completed, the system should present a clear and direct result to the user, present a clear and direct result to the user, such as Compliant / Non-Compliant. The purpose of this feature is to help users quickly understand the result of the request and provide a basis for further manual review or follow-up actions. Compared with relying solely on manual drawing interpretation, a system-based result display makes the workflow more straightforward.
- **Basic Recording and Traceability Support**
  In addition to the basic input and assessment functions, the MVP should also provide initial support for recording key information from the handling process, such as the assessment time, the user who performed the request, the selected asset or structure, the entered load information, and the comparison result. Although a full audit mechanism may not be implemented at this stage, the system should demonstrate the advantage of a digital process in terms of record-keeping and traceability. This would not only improve workflow clarity, but also provide a foundation for stronger compliance and audit support in later stages.

The agreed MVP emphasizes:

- Simplicity and usability
- Clear workflow execution
- Demonstration of system value
- Feasibility within the project timeline

Overall, the MVP agreement reflects a balance between the client's long-term vision and the practical constraints of the project, ensuring that the delivered system is both achievable and aligned with client needs.

2.4 MVP Prioritizations

Based on the kickoff materials, meeting discussions, and the team's current definition of the project scope, the MVP prioritization for this stage is centered on the core workflow of handling structural load requests. The MVP is intended to support the main user roles identified in the project workflow, including System Admin, Asset Manager, and

Contractor. Although the client did not provide a strict ranking of individual features during communication, feature importance was understood in terms of the level of practical need. On this basis, the team identified the functions that most directly support the main user workflow as the core content of the MVP for this stage.

At this stage, basic login and access for authorized users, structure selection, load input, structural limit comparison, and result display form the core features of the MVP. Together, these functions create a complete basic workflow that supports users in moving from request input to result retrieval. In practical terms, this means that relevant users can enter the system, select a structure, input a proposed load, and receive a basic assessment result. Among these functions, structural limit comparison and result assessment are the most critical components, as they directly correspond to the system's main practical purpose: providing a basic assessment of the proposed load.

By comparison, some other functions may still offer practical value, but are closer to features that would further improve the completeness of the system once the core workflow has been established. Examples include more detailed role-based permission management, more advanced engineering calculation logic, a more complete notification mechanism, and broader audit and compliance support. These can be regarded as features that would exceed expectations if achieved, rather than belonging to the core scope of the MVP at this stage.

Therefore, the MVP prioritization for this stage is clearly focused on implementing the main request-handling workflow, namely allowing authorized users in different project roles to enter the system, select a structure, input a load, perform a limit comparison, and view the result.

3. Project planning and management

3.1 Project & Workflow Management Methodology

To ensure structured development and effective collaboration, the team adopts a Kanban-based Agile workflow implemented using GitHub.

Kanban was selected due to its flexibility and suitability for iterative development, allowing tasks to be completed incrementally while maintaining full visibility of progress. This approach is particularly effective for coordinating multiple components of the system, including frontend, backend, and business logic development.

Tasks are managed using:

- **GitHub Issues** for task definition and tracking
- **Kanban Board** for visual progress tracking
- **Milestones** aligned with project deliverables (D1, D2, D3)
- **Pull Requests** with peer review to ensure code quality

The workflow structure is illustrated below:

Figure 1: Kanban Workflow for Task Management

Each task progresses through defined stages, ensuring accountability and transparency across the team.

The team follows a structured communication cadence:

- **Internal Meetings (Wednesdays):** Task allocation, progress tracking, issue resolution
- **Client Meetings (Thursdays):** Requirement validation, feedback, and milestone reviews

This ensures continuous alignment with client expectations and supports iterative refinement.

3.2 Roles and Responsibilities

Responsibilities are assigned based on technical expertise to ensure efficient system development and clear ownership of components.

ROLE	NAME	RESPONSIBILITY
UI/UX Designer	Aoli Wang	Design user interface and user journey, create wireframes and mockups, ensure usability and consistency of the system
Cybersecurity Engineer	Keerthana Narkunaraja	Design secure authentication and access control mechanisms, ensure data protection and confidentiality, identify and mitigate security risks, support secure API and system design
Frontend Engineering Lead	Jiasen Niu	Develop frontend components, integrate UI with backend APIs, manage client-side logic and user interactions
Backend Engineering Lead	Xinghe Wang	Design backend APIs, manage database schema, implement server-side logic and system architecture
DevOps Engineer	Rui Xin	Manage deployment and environment setup, support system integration, maintain version control workflows and CI/CD processes
Business Logic Engineer	Stewie Yang	Develop load comparison engine, implement evaluation logic, manage compliance rules and calculations

Role Alignment with Development Phases

Responsibilities are aligned with system development phases to ensure efficient delivery:

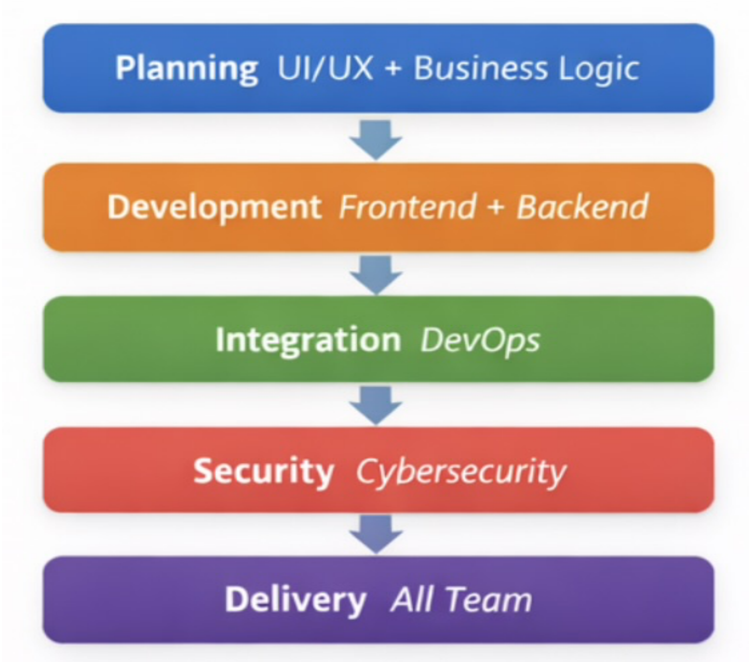


Figure 2: Role Alignment with Development Phases

This ensures that each stage of the system is handled by specialised roles, improving efficiency and reducing development risks.

3.3 Task Breakdown

The project is divided into key task categories to ensure structured development:

- Client communication and requirement validation
- UI/UX design and workflow development
- System architecture and database design
- Backend API and business logic implementation
- Frontend interface development
- Evaluation engine (load comparison logic)
- Email notification system
- System integration and testing

Each task is tracked through GitHub Issues and assigned to responsible team members.

3.4 System Development Workflow

The system is developed based on the user workflow to ensure alignment with functional requirements.

Figure 3: System Development Workflow

This workflow reflects the core functionality of the AssetGuard AI system. Each module is developed independently and integrated progressively.

The modular approach ensures:

- Independent development of features
- Early validation of components
- Reduced integration risks
- Continuous delivery of functional features

3.5 Timeline and Milestones

The project timeline is structured into six phases, aligned with weekly meetings and deliverables.

Phase 1 – Requirements & MVP Confirmation

25 Feb – 12 Mar 2026

- Define user roles and workflows
- Validate MVP with client
- Deliver UI/UX mockups

Phase 2 – System Design & Architecture

12 Mar – 18 Mar 2026

- Database schema design
- Backend architecture (Flask APIs)
- AI integration planning
- Rule engine definition

Phase 3 – Core Feature Development

19 Mar – 15 Apr 2026

- Authentication and role management
- Asset management module
- Evaluation engine
- Email notification system
- Dashboard implementation

Phase 4 – Integration & Testing

16 Apr – 29 Apr 2026

- Frontend and backend integration
- End-to-end testing
- AI data validation

Phase 5 – Refinement & Feedback

30 Apr – 13 May 2026

- UI improvements
- Bug fixing
- Client feedback integration

Phase 6 – Final Delivery

14 May – 21 May 2026

- Final system deployment
- Documentation
- Client demonstration

Milestones

- MVP confirmed → 12 Mar 2026
- D1 submission → 23 Mar 2026
- Prototype ready (D2) → 30 Apr 2026
- Final system delivery (D3) → 21 May 2026

Gantt Chart

Capstone Sprints Spreadsheet.xlsx

The timeline aligns with weekly internal (Wednesday) and client (Thursday) meetings, ensuring continuous validation and progress tracking.

3.6 Collaboration Tools and Practices

The team uses a combination of tools to ensure effective collaboration and project tracking:

- **GitHub** – Code repository, issue tracking, milestones, pull requests
- **Kanban Board** – Visual task tracking
- **Microsoft Teams** – Meetings and communication
- **Excel Gantt Chart** – Timeline planning

Communication Workflow

Figure 4: Communication and Feedback Loop

This structure ensures continuous communication between team members and the client, enabling rapid feedback and iterative improvement.

3.7 Development Approach

The system is developed using a modular and feature-based approach aligned with user workflows, as presented to the client

Each feature (authentication, asset management, evaluation engine, etc.) is built incrementally and validated through weekly client meetings.

This approach ensures:

- Continuous client validation
- Reduced development risk
- Incremental delivery of features

- High system reliability

4. Risk and technology assessments

4.1 Risk Assessment

A number of risks that may impact the progress and quality of the AssetGuard AI project were recognized.

The first significant risk is ambiguity of requirements and change in scope. Some features may be amended or the same features may be refined over the course of the project by the client as their expectations mature. So this could impact the scope and timeline of the project. To reduce this risk, the team will keep communication with the client and facilitator going, maintain clear meeting notes, and keep MVP to only focus on the critical features from day one.

The second risk is integrating with systems. The whole project consists of several connected parts like front end, back end, database, compliance checking logic, and email notifications. The risk is that there can be problems with integration even if each component works well independently. The group will allocate roles, use GitHub for version control, and test the system at regular intervals instead of waiting until the end.

Another significant risk is lack of knowledge of compliance rules in the Engineering domain. Because the system must decide if a load is compliant, the team needs to get to know the rules, thresholds, or practical checking conditions. Without design of compliance logic, it is conceivable that results produced by the system may be incorrect. But to fix that, the team will keep the MVP logic simplistic, test assumptions with the client, and start out with only a handful of clear and manageable compliance cases.

Time management and workload balancing are risk factors as well. Since this is part of group work team members might also have various schedules, other studies, or experience with some technologies being chosen. This means some responsibilities may take longer than anticipated. This risk will be managed by assigning specific responsibilities to each member, meeting weekly, and then reflecting on the progress, which can help in finding delays early.

The artificial intelligence component is also unclear. AI tools might entail concerns about cost, quality of output, veracity, reliability, and end-user experience. Thus for this reason, the group will not make compliance decisions entirely with the help of AI. Instead, we will have a stringent checking process in place in terms of compliance logic and AI will only be used in the form of clearer explanations, alerting warnings, and providing better user communications.

Last but not least, low quality of testing could be influenced by small sample size. Limited real-world examples, if any, could limit fully assessing the value of a system. To minimise this risk the group will use the sample data it has to be a starting point, and then will add more test cases where relevant, and design the system to allow more complete numbers of cases to be added (or to be added at a later stage) without much re-design.

4.2 Skills and Resources Analysis

The strengths on this team are diverse and ready at different levels. Some members of the team know front-end development (React). Some other members of the team are better at back-end development, integration of apps with APIs and data. This diversity should help lead to an established web-based compliance portal as well.

Several areas needed further learning within the team were also identified. Although the core members will have basic programming and web development experience, some of the students are new to Flask back-end development and SQLite database integration (in their opinion) and are not as well versed in using AI APIs in a live application as others. Besides, given how the project is concerned with testing engineering compliance the team needs time to understand the relevant rules and how they should be carried out differently in the computer in our system.

In terms of resources, members of the team can work on meetings/development/testing / documentation every week. The team is also able to use personal laptops, the required software development tools, and university portals like GitHub and Microsoft Teams. So we have enough resources to achieve the size and scope of the proposed MVP.

With the goals to improve the gaps shown in skills identified, the team intends on taking steps to learn new technologies and tools during the course of working on them (in official documentation & lab testing etc.) and sharing knowledge within our organization. This involves learning Flask and SQLite and the chosen AI service too in order to have access to those technologies as a first and second skill acquisition. In summary. As described previously they believe that whatever skills and resources must be had for the MVP we are going to do so it can get done and tasks should remain more precise.

4.3 Technology Assessment

For that project, the team chose technology stack, considering requirements of the project, client requirements as well as MVP scope and what is realistically achievable in a semester.

The project is being developed as a web site portal instead of desktop application. A web one was suggested for this, it is more convenient to use, easy to access by users on various devices and OSs, and it can be easily read over the network without having to worry about installing and updating it locally. This approach is a cost-effective one for the client and lessens maintenance burden in the future.

To implement the front end, the team will use React, combined with Tailwind CSS along with shadcn/ui. A clean, responsive interface for feature development (e.g., login, asset selection, load input, compliance result display), should benefit from this combination. React is component-based development so a portal would have several interactive pages to manage using this type of framework while Tailwind CSS and shadcn/ui help in making interface development faster and maintain visual consistency.

The main database the team is going to use is SQLite. This is not surprising as most work on this project focuses at user information, infrastructure details and user submission lists, compliance records and emails. Therefore, a lightweight relational database will do best for the MVP to be in place for easy usage and easy to use.

Appendix

1. Related Links

- GitHub repository: https://github.com/xinghe0709/AssetAI-Guard
- GitHub Kanban : https://github.com/users/xinghe0709/projects/2
- Teams Folder: https://uniwa.sharepoint.com/teams/InformationTechnologyCapstoneProjectSEM-12026-Group11/Shared%20Documents/Forms/AllItems.aspx

2. MVP Scope Confirmation & Client Q&A Record

Date: 19 March 2026, 2:00 PM – 2:30 PM

Client Representatives: Glen Pikor, Ryan Morrison

Team Representatives: Group 11

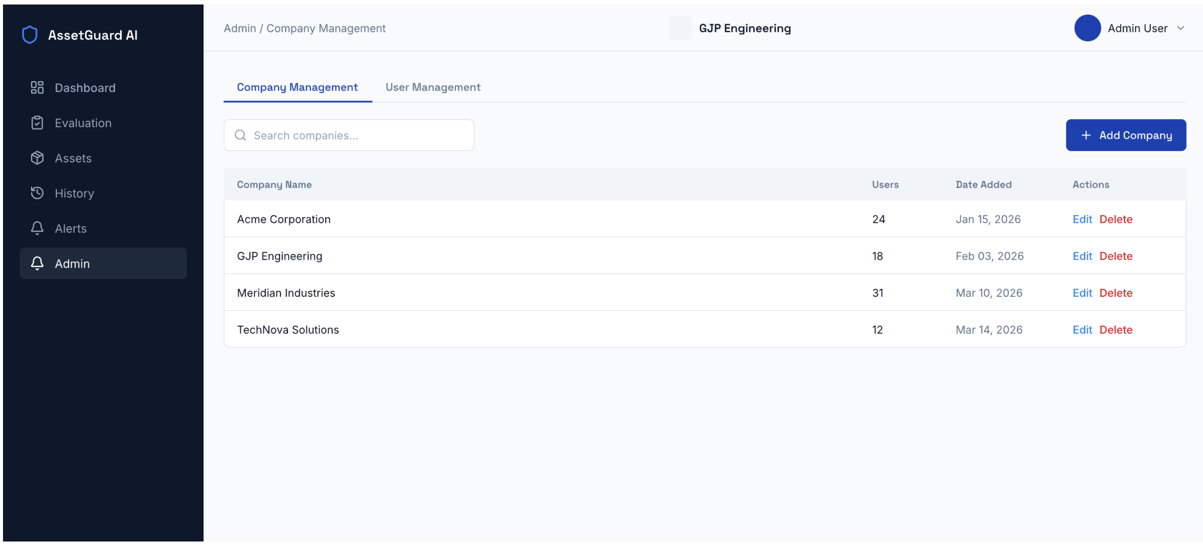
Q (Team)	Regarding user roles, we have defined "Admin" and "Asset Manager." Could you clarify the specific use case for the "Contractor" role mentioned previously?
A (Client)	Contractors are typically personnel like crane operators who need to apply a specific load (e.g., a 200kN point load) onto an asset and need to know if it is safe to do so.
Q (Team)	Should Contractors have the ability to modify asset data?
A (Client)	No. Contractors can perform evaluations and view results, but they should not have access to adding or deleting assets. That is the role of the Asset Manager.
Q (Team)	MVP would not include features like "Self-signup" or "Forgot Password", would this be acceptable?
A (Client)	Yes. For the MVP, it is fine for the System Admin to handle all user account creation and management.
Q (Team)	We noticed both "Structures" and "Assets" used in the vision. Should we standardize this?

A (Client)	Yes, let's standardize everything as "Assets." While a structure specifically refers to something that transfers a load, all structures are assets in this system.
Q (Team)	The system will support both manual entry by the manager and batch imports to integrate with our existing data structures. Would this align with your expectation?
A (Client)	Yes, we will provide your some sample data and sample drawings.
Q (Team)	Is the evaluation logic limited to numeric comparisons and unit conversions for the MVP?
A (Client)	Yes, perform numeric comparisons against design limits and take unit conversions into account.
Q (Team)	Are there any other fields required for the evaluation input?
A (Client)	Please add a "Comments" field. Contractors might need to specify details, such as using spreader plates under their crane outriggers, which is critical for the final engineering check.
Q (Team)	We suggest to send emails by user-triggered, instead of automatically, to avoid spam emails, Does this make sense for this situation?
A (Client)	Let the user manually trigger the email. However, the most important thing is that every evaluation attempt is saved in a persistent database for traceability, whether an email is sent or not.
Q (Team)	Who should receive the assessment reports?
A (Client)	The report should be sent to the user (contractor) and also be accessible to the Asset Manager so they have a formal record of the communication and the safety result.
Q (Team)	We have prioritized the modules as: 1. Auth & Roles, 2. Asset Management, 3. Evaluation Engine, 4 Email System., 5. Dashboard. Does this align with your needs?
A (Client)	Please prioritize the dashboard before the email system. The Dashboard is great for visualizing historical data. If you run out of time, the external email system is less critical than the internal logging and comparison engine.
Q (Team)	We are proceeding with a tech stack of React (Frontend), Flask (Backend), and SQLite (Database).
A (Client)	Agreed. As long as it is a web-based portal that is easy to access across different devices.

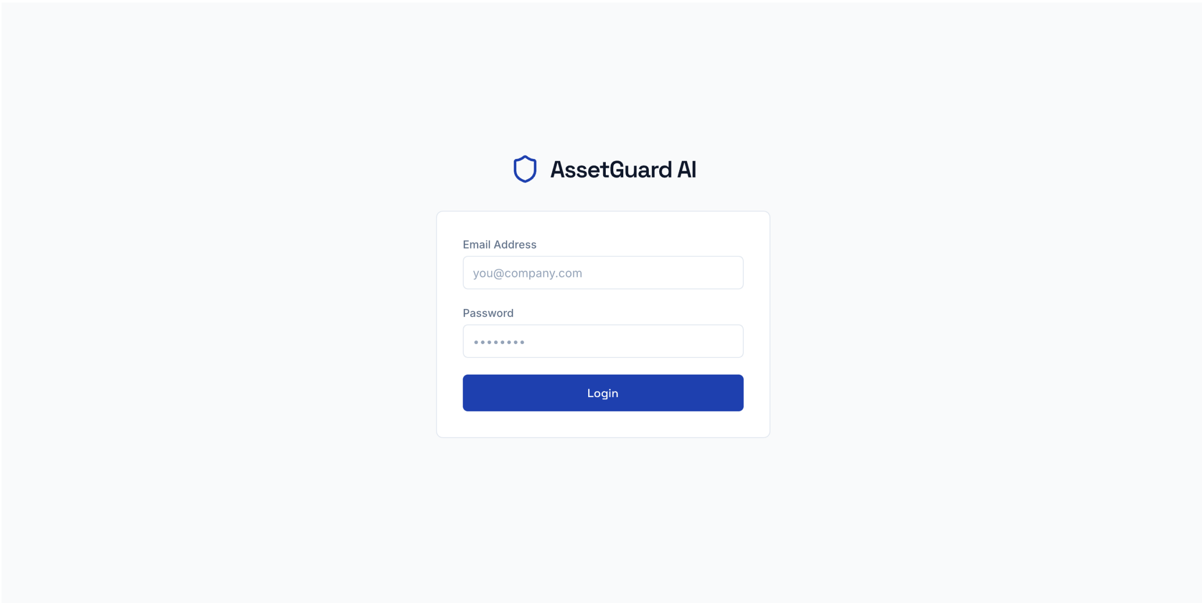
3. Work-in-Progress (WIP) UX Framework

The following wireframes represent the initial logic flows and structural frameworks presented to GJP Engineering during the Week 4 meeting. **Please note that these are not final UI designs but rather "work-in-progress" (WIP) visualizations** used to validate core user journeys and functional requirements with the client. In the upcoming development phase, the team will further refine these into high-fidelity user interface, incorporating GJP branding.

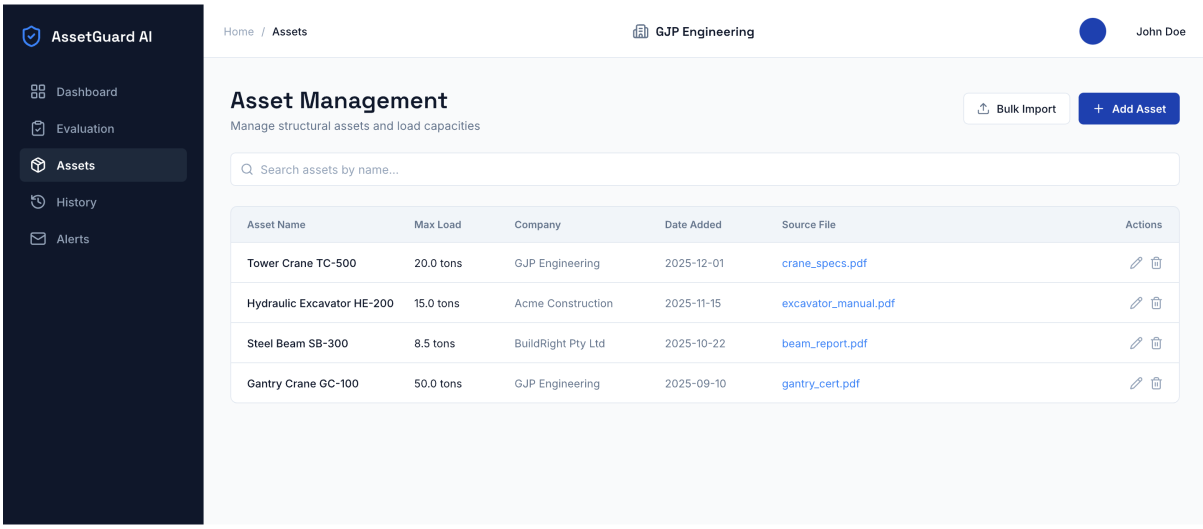
- Admin portal – create user



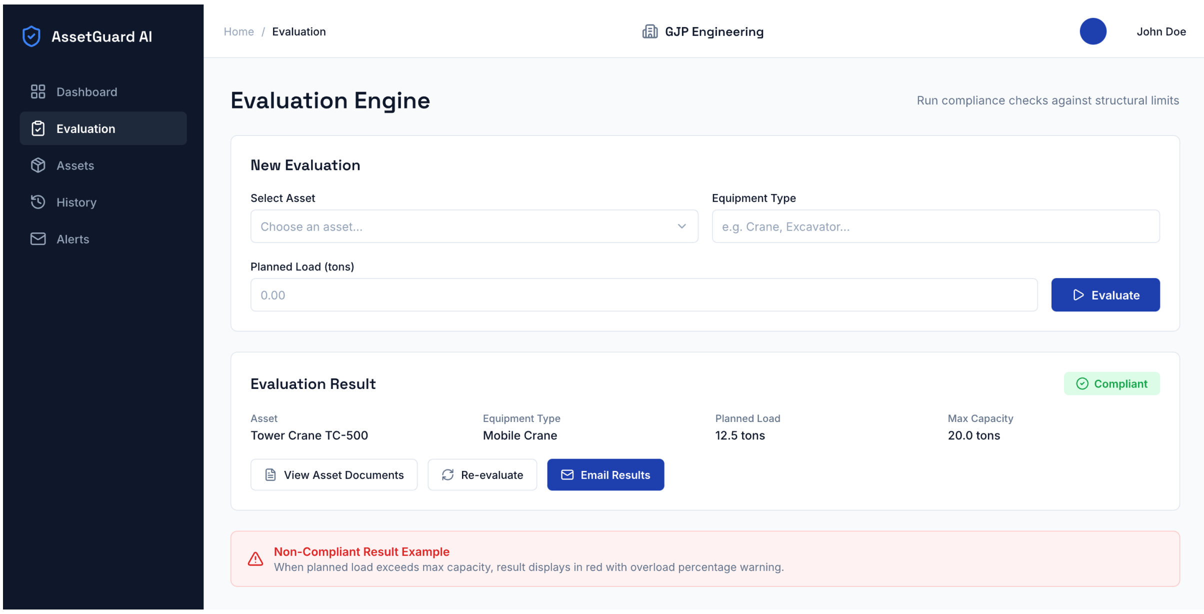
- User login



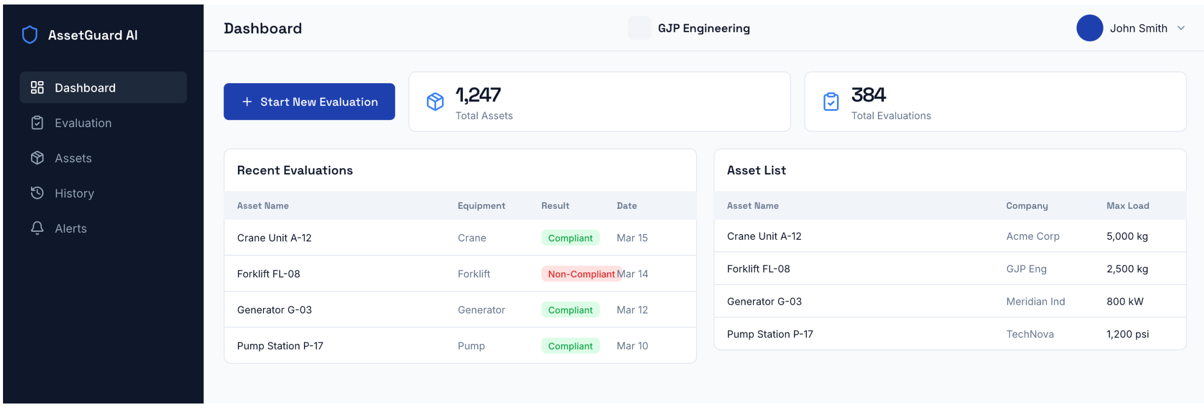
- Asset Management



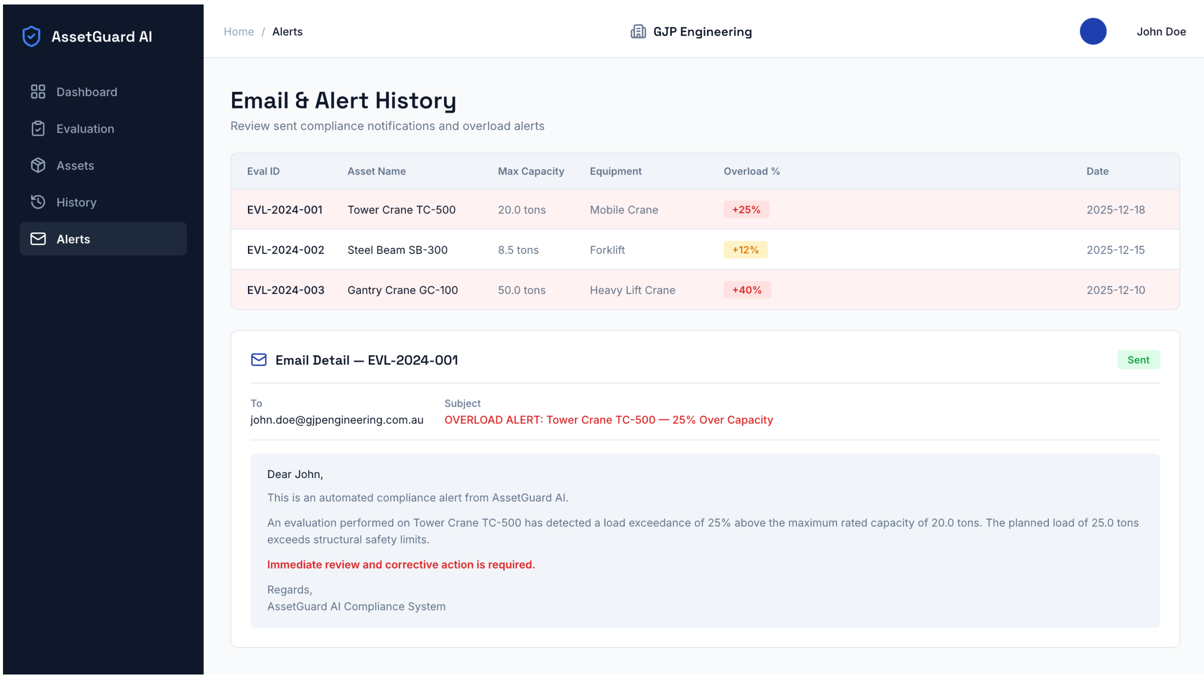
- Evaluation



- Dashboard



- Email



Disclosure: AI assistance (ChatGPT/Gemini) was used to aid understanding of project requirements and improve the clarity of expression in parts of this report. All outputs were critically reviewed and revised by the authors to ensure academic integrity.